

Market Size Weights and Outside Option: Technical Notes

Overview

This document addresses three questions about the weighting scheme used across the sleepiness and BLP demand estimation pipelines in the Egan et al. (2025) replication:

1. What are the appropriate weights for aggregating the sleepiness function $\hat{\phi}_{mt}$ to the national level?
2. How does the demand estimation handle market size M_{mt} for D-firms and the outside option, given that "active deposit market size" is only well-defined for B-firms?
3. Why is the γ -ceiling used to compute the outside option share innocuous?

1. Sleepiness Aggregation Weights

The Aggregation Problem

The sleepiness estimation produces a local estimate $\hat{\phi}_{mt}$ for each market m at time t . To compute the national aggregate $\hat{\phi}_t$ (used for D-firms in the demand prep), we need a weighted average:

$$\hat{\phi}_t = \frac{\sum_m M_{mt} \cdot \hat{\phi}_{mt}}{\sum_m M_{mt}}$$

The question is: what is M_{mt} ?

Why Population Is Correct

Under the model assumption that local deposit market potential is a constant fraction of population:

$$M_{mt} = c \cdot \text{pop}_{mt}$$

the constant c cancels from the numerator and denominator:

$$\hat{\phi}_t = \frac{\sum_m c \cdot \text{pop}_{mt} \cdot \hat{\phi}_{mt}}{\sum_m c \cdot \text{pop}_{mt}} = \frac{\sum_m \text{pop}_{mt} \cdot \hat{\phi}_{mt}}{\sum_m \text{pop}_{mt}}$$

This makes `pop_total` the correct weight. Using lagged deposits instead would introduce an endogeneity concern: markets with high deposits (partly *because* of low sleepiness) receive disproportionate weight, biasing the national aggregate downward.

Implementation

The `calculate_phis()` and `calculate_pooled_phis()` functions in all five sleepiness scripts (`estimation_{1,3,4,5,6}_sleep.py`) now set:

```
if 'pop_total' in df.columns:
    df['market_size'] = df['pop_total'].fillna(0)
else:
    df['market_size'] = 1.0
```

This weight enters the aggregation:

```
market_agg = df.groupby(['year_quarter', 'CODMUN_IBGE']).agg(
    phi_mt=('phi_mt_Spec', 'mean'),
    M_mt=('market_size', 'sum')
)
national_phi_t = Σ(phi_mt * M_mt) / Σ(M_mt)
```

The demand prep script (`estimation_1_demand_1_prep.py`) already uses `pop_total` for the same aggregation:

```
def weighted_mean(g):
    v, w = g['phi_mt'], g['pop_total']
    return v.mean() if w.sum() == 0 else np.average(v, weights=w)
```

So the two stages are now consistent.

Scripts Modified

Script	Function	Previous Weight	New Weight
<code>estimation_1_sleep.py</code>	<code>calculate_phis()</code>	<code>lagged_deposits</code>	<code>pop_total</code>

Script	Function	Previous Weight	New Weight
estimation_3_sleep.py	calculate_local_phis() , plotting	lagged_deposits	pop_total
estimation_4_sleep.py	calculate_pooled_phis()	lagged_deposits	pop_total
estimation_5_sleep.py	calculate_pooled_phis()	lagged_deposits	pop_total
estimation_6_sleep.py	calculate_pooled_phis()	lagged_deposits	pop_total

2. Market Size in the Demand Estimation

The Identification Challenge

In the BLP demand estimation, the "market size" M_{mt} defines the potential number of consumers. Shares are then:

$$s_{jkmt} = \frac{q_{jkmt}}{M_{mt}}$$

where q_{jkmt} is a quantity (active deposits). The market size must satisfy $\sum_j q_j < M_{mt}$ so that the outside option share $s_0 > 0$.

For **B-firms**, market size is straightforward: $M_{mt}^B = \gamma \cdot \text{pop}_{mt}$, where γ is calibrated from the data.

For **D-firms** and the **outside option**, there is no natural market-level quantity — D-firms are national, and the outside option has no observed deposits at all.

How We Handle It

The demand prep script (estimation_1_demand_1_prep.py) calibrates a single global per-capita ceiling γ :

$$\gamma = 1.1 \times \max_{m,t} \left(\frac{\sum_j \text{Dep}_{jmt}^{\text{Act}}}{\text{pop}_{mt}} \right)$$

This defines:

- **B-firm shares:** $s_{jkm t}^B = \text{Dep}_{jkm t}^{\text{Act}} / (\gamma \cdot \text{pop}_{m t})$
- **D-firm shares:** $s_{jkt}^D = \text{Dep}_{jkt}^{\text{Act}} / (\gamma \cdot \text{Pop}_t^{\text{National}})$

Both use the same γ multiplied by population. The outside option share in market m is implicitly:

$$s_{0m t} = 1 - \sum_{j \in J_{m t}^B} s_{j m t}^B - \sum_{j \in J_t^D} s_{j m t}^{D, \text{local}}$$

Inside the BLP loop (`compute_model_shares`), the D-firm national share is aggregated using population weights:

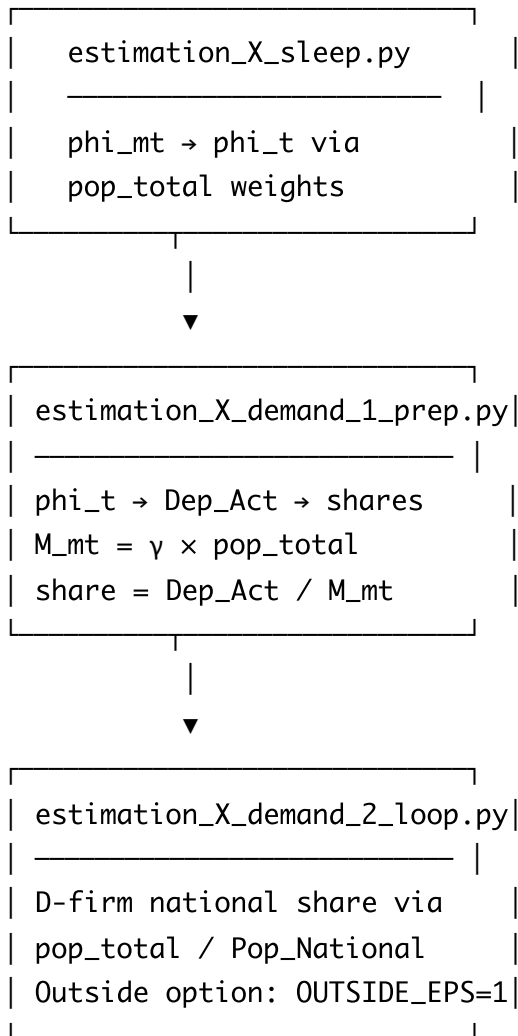
$$s_{jkt}^{D, \text{National}} = \sum_m \frac{\text{pop}_{m t}}{\text{Pop}_t} \cdot \frac{\exp(V_{jkt})}{\text{Denom}_{m t}}$$

This is model-consistent: since $M_{m t} = \gamma \cdot \text{pop}_{m t}$, the weight ratio is:

$$\frac{M_{m t}}{M_t} = \frac{\gamma \cdot \text{pop}_{m t}}{\gamma \cdot \text{Pop}_t} = \frac{\text{pop}_{m t}}{\text{Pop}_t}$$

The γ cancels. So population weights are the *only* internally consistent choice. Using active deposit totals as weights would be circular (deposits depend on shares, which depend on the weights).

Summary of Weight Flow



All three stages are internally consistent under the assumption $M_{mt} = c \cdot \text{pop}_{mt}$.

3. Why the γ -Ceiling Is Innocuous

The ceiling γ is computed as:

```
gamma = dep_per_capita.max() * 1.1
```

where `dep_per_capita` = total active deposits in market (m, t) divided by `pop_total`. Three reasons this is harmless:

3.1 γ Drops Out of the Estimation

Inside the BLP loop, the contraction mapping operates on:

$$T(\delta)_j = \delta_j + \ln s_j^{\text{data}} - \ln s_j^{\text{model}}$$

Since $s_j^{\text{data}} = \text{Dep}_j^{\text{Act}} / (\gamma \cdot \text{pop}_{mt})$, changing γ shifts *all* data shares by $1/\gamma$. In log-space, this is $\ln s^{\text{data}} = \ln \text{Dep}^{\text{Act}} - \ln \gamma - \ln \text{pop}$. The $-\ln \gamma$ term shifts all δ_j by a common constant, which is absorbed into the intercept of θ_1 (the FGC dummy or the overall constant in the linear IV step). The structural parameters θ_2 (random coefficients), the relative α_k (deposit-type spread coefficients), and all β (product characteristic coefficients) are **invariant to γ** .

More precisely: if we replace γ with $\gamma' = a \cdot \gamma$ for any $a > 0$, then $s_j^{\text{data}'} = s_j^{\text{data}}/a$ and $\delta'_j = \delta_j - \ln a$ for all j . Since θ_1 is recovered from $\delta = X\theta_1 + \xi$, the shift $-\ln a$ is absorbed into whichever column of X acts as an intercept. The residuals ξ are unchanged, so the GMM moments and the entire θ_2 estimation are unaffected.

3.2 The 10% Padding Guarantees $s_0 > 0$

By construction, the densest market has $\sum_j s_j \leq \gamma^{-1} \cdot \max_{m,t} (\sum_j \text{Dep}_j^{\text{Act}} / \text{pop}_{mt}) = 1/1.1 \approx 0.91$. This guarantees at least a 9% outside option share in every market. Without this padding, the densest market could have $\sum_j s_j \approx 1$, causing:

- $\ln s_0 \rightarrow -\infty$, destabilizing the contraction.
- The outside option failing to pin the level of δ , reintroducing shift-invariance.

3.3 The Outside Option Level Is Not Separately Identified

BLP identifies utility *differences* relative to the outside good. Changing γ changes the implied s_0 and shifts all δ_j uniformly, which is isomorphic to changing the intercept. Since no utility parameter is estimated for the outside option (its utility is normalized to $V_0 = 0$), the absolute level of γ is irrelevant for every identified parameter in the model.

This is a standard result: see Berry (1994, Section 2), where the normalization $u_{i0t} = 0$ makes the outside option share a function of the data transformation $(\ln s_j - \ln s_0)$, not a structural parameter. The γ -ceiling simply determines what s_0 is, which affects the level of δ but not the parameters that the econometrician cares about.

References

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